Untyped Lambda Calculus

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Chapter 1

Alpha equivalence equalities

Definition 1 (Agreement set)**.** For permutations π, π' of the set of atoms $\mathbb{A}$, the *agreement set* is

$$AS(\pi, \pi') \stackrel{\text{def}}{=} \{\, a \in \mathbb{A} \mid \pi(a) = \pi'(a) \,\}.$$

([Crole2012], Section 2.)

Definition 2 (Disagreement set)**.** For permutations π, π' of the set of atoms $\mathbb{A}$, the *disagreement set* is

$$DS(\pi, \pi') \stackrel{\text{def}}{=} \{\, a \in \mathbb{A} \mid \pi(a) \neq \pi'(a) \,\}.$$

([Crole2012], Section 2.)

Definition 3 (Permutation action on expressions)**.** The action $\pi \cdot E$ of a (finitely supported) permutation π of atoms on an expression E is defined by recursion on E:

$$\pi \cdot a \stackrel{\text{def}}{=} \pi(a), \qquad \pi \cdot P(E_1, E_2) \stackrel{\text{def}}{=} P(\pi \cdot E_1, \pi \cdot E_2), \qquad \pi \cdot B([a]E) \stackrel{\text{def}}{=} B([\pi \cdot a]\, \pi \cdot E).$$

([Crole2012], Section 2.)

Definition 4 (Atom swapping)**.** For atoms x, y, the transposition $(x\, y)$ is the permutation exchanging x and y and fixing every other atom; its action $(x\, y) \cdot E$ on an expression E is the special case of theorem 3 in which π is a transposition, so that $(x\, y) \cdot E$ is E with all occurrences of x and y swapped. ([Crole2012], Section 2; [Gabbay2002], Section 2.)

Definition 5 (Definition 3.2: the relation $\sim_{p\#}$)**.** The binary relation $\sim_{p\#}$ on the set of expressions is inductively defined by

$$\frac{}{a \sim_{p\#} a}\ atom \qquad \frac{E_1 \sim_{p\#} E_1' \quad E_2 \sim_{p\#} E_2'}{P(E_1, E_2) \sim_{p\#} P(E_1', E_2')}\ pcg \qquad \frac{(z\, a) \cdot E \sim_{p\#} (z\, b) \cdot E'}{B([a]E) \sim_{p\#} B([b]E')}\ \pi\# \quad [\, z \# a, b, E, E' \,]$$

where the side condition $z \# a, b, E, E'$ means that z differs from a and b and has no free occurrence in E or E'. ([Crole2012], Definition 3.2.)

Definition 6 (Definition 3.3: the relation $\sim_p^1$)**.** The binary relation $\sim_p^1$ on expressions is defined by the rules of theorem 12, but with the rule π replaced by

$$\frac{(z\, a) \cdot E \sim_p^1 (z\, b) \cdot E'}{B([a]E) \sim_p^1 B([b]E')}\ \pi_1 \quad [\, z \not\triangleright E, E' \,]$$

that is, the witness z is only required not to occur in the two bodies. ([Crole2012], Definition 3.3.)

Definition 7 (The relation $\sim_p^2$ of Proposition 4.3)**.** The binary relation $\sim_p^2$ on expressions is defined by the rules of theorem 12, but with the rule π replaced by

$$\frac{(z\,a)\cdot E \sim_p^2 (z\,b)\cdot E'}{B([a]E) \sim_p^2 B([b]E')}\ \pi_2 \quad [\,z \not\approx a, b\,]$$

that is, the witness z need only differ from the two binding atoms. ([Crole2012], Proposition 4.3.)

Definition 8 (The relation $\sim_p^3$ of Proposition 4.3)**.** The binary relation $\sim_p^3$ on expressions is defined by the rules of theorem 12, but with the rule π replaced by

$$\frac{(z\,a)\cdot E \sim_p^3 (z\,b)\cdot E'}{B([a]E) \sim_p^3 B([b]E')}\ \pi_3$$

which carries no side condition at all. ([Crole2012], Proposition 4.3.)

Definition 9 (Definition 3.4: the relation $\sim_r$)**.** The binary relation $\sim_r$ on expressions is inductively defined by

$$\frac{}{E \sim_r E}\ \mathit{ref} \qquad \frac{E_1 \sim_r E_2}{E_2 \sim_r E_1}\ \mathit{sym} \qquad \frac{E_1 \sim_r E_2 \quad E_2 \sim_r E_3}{E_1 \sim_r E_3}\ \mathit{trs}$$

$$\frac{E_1 \sim_r E_1' \quad E_2 \sim_r E_2'}{P(E_1, E_2) \sim_r P(E_1', E_2')}\ \mathit{pcg} \qquad \frac{E \sim_r E'}{B([a]E) \sim_r B([a]E')}\ \mathit{bcg}$$

$$\frac{}{B([a]E) \sim_r B([a']E\{a'/a\})}\ \alpha \quad [\,a' \not\approx a, E\,]$$

where $E\{a'/a\}$ is capture-avoiding substitution and the side condition $a' \not\approx a, E$ says that a' differs from a and does not occur at all in E. ([Crole2012], Definition 3.4.)

Definition 10 (Definition 3.5: the relation $\sim_{r\#}$)**.** The binary relation $\sim_{r\#}$ on expressions is defined by the rules of theorem 9, but with the rule α replaced by

$$\frac{}{B([a]E) \sim_{r\#} B([a']E\{a'/a\})}\ \alpha\# \quad [\,a' \,\#\, a, E\,]$$

whose side condition only requires a' to differ from a and to have no *free* occurrence in E. ([Crole2012], Definition 3.5.)

Definition 11 (Naive renaming)**.** For an expression E and atoms a, z, the expression $E[a \mapsto z]$ is E with every occurrence of a, whether free, bound or binding, replaced by z. This is not an operation of [Crole2012]. If z does not occur in E, it coincides with the swap $(z\,a)\cdot E$ of theorem 4.

Definition 12 (Definition 3.1: the relation $\sim_p$)**.** The binary relation $\sim_p$ on the set of expressions is inductively defined by

$$\frac{}{a \sim_p a}\ \mathit{atom} \qquad \frac{E_1 \sim_p E_1' \quad E_2 \sim_p E_2'}{P(E_1, E_2) \sim_p P(E_1', E_2')}\ \mathit{pcg} \qquad \frac{(z\,a)\cdot E \sim_p (z\,b)\cdot E'}{B([a]E) \sim_p B([b]E')}\ \pi \quad [\,z \not\approx a, b, E, E'\,]$$

where the side condition $z \not\approx a, b, E, E'$ means that z differs from a and b and does not occur in E or E'. The formalization states the premise of π with the renaming of theorem 11, which agrees with the swap under this side condition. ([Crole2012], Definition 3.1.)

Lemma 13 ($\sim_p$ is symmetric)**.** *For all expressions E, E' we have $E \sim_p E' \implies E' \sim_p E$. ([Crole2012], proof of Theorem 4.4, case* sym*.)*

Proof. □

Lemma 14 (Lemma 6.1: transpositions preserve $\sim_p$)**.** *For any atoms u and v and expressions E and E' we have*

$$E \sim_p E' \implies (u\,v) \cdot E \sim_p (u\,v) \cdot E'.$$

([Crole2012], Lemma 6.1.)

Proof. □

Lemma 15 ($\sim_p$ is reflexive)**.** *For every expression E we have $E \sim_p E$. ([Crole2012], proof of Theorem 4.4, case* ref*.)*

Proof. □

Lemma 16 ($\sim_p$ is closed under the rule bcg)**.** *For all expressions E, E' and every atom a we have $E \sim_p E' \implies B([a]E) \sim_p B([a]E')$. ([Crole2012], proof of Theorem 4.4, case* bcg*.)*

Proof. □

Lemma 17 ($\sim_p$ is transitive)**.** *For all expressions E, E', E'' we have $E \sim_p E' \sim_p E'' \implies E \sim_p E''$. ([Crole2012], proof of Theorem 4.4, case* trs*.)*

Proof. □

Lemma 18 (Lemma 6.7: substitution preserves α-equivalence)**.** *For any expressions E, E', R, R' and any atom a,*

$$E \sim_p E' \text{ and } R \sim_p R' \implies E\{R/a\} \sim_p E'\{R'/a\}.$$

This is Lemma 6.7 of [Crole2012], which is stated there for $\sim_r$ and for the renaming $E\{a/b\}$; by Theorem 4.4 the two statements agree, and the version formalised here allows an arbitrary expression to be substituted, and its two sides to differ up to α-equivalence.

Proof. □

1.1 Lemmas on expressions

The lemmas of Section 6.1 of [Crole2012], which support the proofs of the equalities of Section 4.

Lemma 19 (Lemma 6.2(1): permutations agreeing on occurring atoms)**.** *For any expression E and permutations π, π',*

$$\mathrm{occ}(E) \subseteq AS(\pi, \pi') \implies \pi \cdot E = \pi' \cdot E.$$

([Crole2012], Lemma 6.2 part 1.)

Proof. □

Lemma 20 (Lemma 6.2(1) for transpositions)**.** *For any expression E and atoms a, u, z with $u \not\triangleright E$ and $z \not\triangleright E$,*

$$(z\,u) \cdot (u\,a) \cdot E = (z\,a) \cdot E.$$

This is the instance of theorem 19 used in [Crole2012] in the proofs of Theorems 4.2 and 4.4.

Proof. □

Lemma 21 (Lemma 6.2(2): permutations agreeing on free atoms)**.** *For any expression E and permutations π, π',*

$$\mathrm{free}(E) \subseteq AS(\pi, \pi') \implies \pi \cdot E \sim_p \pi' \cdot E.$$

([Crole2012], Lemma 6.2 part 2.)

Proof. □

Lemma 22 (Lemma 6.2(2) for transpositions)**.** *For any expression E and atoms a, u, z with $u \# E$ and $z \# E$,*

$$(z\,u) \cdot (u\,a) \cdot E \sim_p (z\,a) \cdot E.$$

This is the instance of theorem 21 used in [Crole2012] in the proof of Theorem 4.1.

Proof. □

Lemma 23 (Lemma 6.3, in naive-renaming form)**.** *For any atoms a, z and any expression E with $z \not\triangleright E$,*

$$E[a \mapsto z] \sim_r E\{z/a\}$$

where $E[a \mapsto z]$ is the naive renaming of theorem 11. This is the form in which the induction of Lemma 6.3 of [Crole2012] is carried out here.

Proof. □

Lemma 24 (Lemma 6.3: swapping versus substitution, for $\sim_r$)**.** *For any atoms a and z and any expression E,*

$$z \not\triangleright E \implies (z\,a) \cdot E \sim_r E\{z/a\}.$$

([Crole2012], Lemma 6.3.)

Proof. □

Lemma 25 (Lemma 6.4, in naive-renaming form)**.** *For any atoms a, z and any expression E with $z \not\triangleright E$,*

$$E[a \mapsto z] \sim_{r\#} E\{z/a\}.$$

This is theorem 23 with every use of the rule α replaced by $\alpha\#$.

Proof. □

Lemma 26 (Lemma 6.4: swapping versus substitution, for $\sim_{r\#}$)**.** *For any atoms a and z and any expression E,*

$$z \not\triangleright E \implies (z\,a) \cdot E \sim_{r\#} E\{z/a\}.$$

([Crole2012], Lemma 6.4.)

Proof. □

Lemma 27 (Lemma 6.6: avoiding an atom, for $\sim_r$)**.** *For all expressions E and atoms a,*

$$a \# E \implies (\exists \hat{E})\,(a \not\triangleright \hat{E} \wedge \hat{E} \sim_r E).$$

Informally, bound atoms can always be renamed so that a particular atom does not occur; this may be regarded as a form of Barendregt's variable convention. ([Crole2012], Lemma 6.6.)

Proof. □

Lemma 28 (Lemma 6.6 for $\sim_p$)**.** *For all expressions E and atoms a,*

$$a \# E \implies (\exists \hat{E})\,(a \not\triangleright \hat{E} \wedge \hat{E} \sim_p E).$$

This is theorem 27 with $\sim_r$ replaced by $\sim_p$. It cannot be obtained from theorem 27 through Theorem 4.4, because the proof of that theorem uses this statement.

Proof. □

Lemma 29 (Lemma 6.3 for $\sim_p$, in naive-renaming form)**.** *For any atoms a, a' and any expression E with $a' \not\triangleright E$,*

$$E[a \mapsto a'] \sim_p E\{a'/a\}.$$

This is theorem 23 with $\sim_r$ replaced by $\sim_p$.

Proof. □

Lemma 30 (Substituting a fresh atom is a transposition)**.** *For any expression E and atoms a, a' with $a' \# E$,*

$$E\{a'/a\} \sim_p (a\,a') \cdot E.$$

This is the first of the two ingredients of theorem 35: note that a' is only required to be fresh for E, bound occurrences being renamed away first using theorem 28.

Proof. □

Lemma 31 (Splitting a transposition through a fresh atom)**.** *For any expression E and atoms z, a, a' with $z \# E$ and $a' \# E$,*

$$(z\,a) \cdot E \sim_p (z\,a') \cdot (a\,a') \cdot E.$$

This is the second ingredient of theorem 35: the two permutations agree on $\mathrm{free}(E)$*, so theorem 21 applies.*

Proof. □

Definition 32 (Chain of transpositions)**.** Given a finite list of triples of atoms $(z_1, a_1, a'_1), \ldots, (z_n, a_n, a'_n)$ and an expression E, the expression

$$(z_1\,a_1) \cdot \ldots \cdot (z_n\,a_n) \cdot E$$

which is the left-hand side of theorem 35.

Definition 33 (Chain of transpositions after a chain of substitutions)**.** Given a finite list of triples of atoms $(z_1, a_1, a'_1), \ldots, (z_n, a_n, a'_n)$ and an expression E, the expression

$$(z_1\,a'_1) \cdot \ldots \cdot (z_n\,a'_n) \cdot E\{a'_1/a_1\} \ldots \{a'_n/a_n\}$$

which is the right-hand side of theorem 35.

Definition 34 (Atoms of a chain)**.** The list of all atoms z_i, a_i, a'_i mentioned by a list of triples $(z_1, a_1, a'_1), \ldots, (z_n, a_n, a'_n)$. Its being repetition-free is the pairwise-disjointness hypothesis of theorem 35.

Lemma 35 (Lemma 6.5: chains of transpositions versus chains of substitutions)**.** *Let the atoms z_i, a_i, a'_i be pairwise disjoint across all i. Then for every expression E with $z_i \# E$ and $a'_i \# E$ for all i,*

$$(z_1\, a_1) \cdot \ldots \cdot (z_n\, a_n) \cdot E \sim_p (z_1\, a'_1) \cdot \ldots \cdot (z_n\, a'_n) \cdot E\{a'_1/a_1\} \ldots \{a'_n/a_n\}.$$

([Crole2012], Lemma 6.5.) The hypotheses $z_i \not\triangleright E$ of the paper are weakened here to $z_i \# E$, which suffices because both ingredients of the proof are invariant under $\sim_p$ and mention only free atoms.

Proof. □

Lemma 36 (Lemma 6.5 for a single triple of atoms)**.** *For any expression E and atoms z, a, a' with $z \not\triangleright E$, $z \neq a$, $a \neq a'$ and $a' \# E$,*

$$(z\, a) \cdot E \sim_p (z\, a') \cdot E\{a'/a\}.$$

This is the case $n = 1$ of theorem 35, which is the step carried out in [Crole2012] inside the proof of Theorem 4.4 for the rule α.

Proof. □

1.2 The equality of the definitions

The five relations of Section 3 of [Crole2012] are all the same relation, each capturing α-equivalence for expressions.

Lemma 37 ($\sim_p$ is contained in $\sim_{p\#}$)**.** *For all expressions E, E' we have $E \sim_p E' \implies E \sim_{p\#} E'$. This is the trivial half of Theorem 4.1 of [Crole2012]: non-occurrence implies freshness.*

Proof. □

Lemma 38 ($\sim_{p\#}$ is contained in $\sim_p$)**.** *For all expressions E, E' we have $E \sim_{p\#} E' \implies E \sim_p E'$. Equivalently, $\sim_p$ is closed under the rules for $\sim_{p\#}$; the only interesting case is the rule $\pi\#$, which is handled by picking a fully non-occurring witness and appealing to theorem 14 and theorem 21. ([Crole2012], Theorem 4.1.)*

Proof. □

Theorem 39 (Theorem 4.1: $\sim_p$ equals $\sim_{p\#}$)**.** *The relation $\sim_p$ of α-equivalence according to theorem 12 is identical to the relation $\sim_{p\#}$ of theorem 5. ([Crole2012], Theorem 4.1.)*

Proof. □

Lemma 40 ($\sim_p$ is contained in $\sim_p^1$)**.** *For all expressions E, E' we have $E \sim_p E' \implies E \sim_p^1 E'$, since the side condition of π is stronger than that of π_1. ([Crole2012], Theorem 4.2.)*

Proof. □

Lemma 41 (The degenerate case of the rule π_1)**.** *For atoms a, b and expressions E, E' with $a \not\triangleright E'$,*

$$E \sim_p (a\, b) \cdot E' \implies B([a]E) \sim_p B([b]E').$$

This is the case $u = a$ in the proof of Theorem 4.2 of [Crole2012], which is settled using theorem 14 and theorem 19.

Proof. □

Lemma 42 ($\sim_p^1$ is contained in $\sim_p$)**.** *For all expressions E, E' we have $E \sim_p^1 E' \implies E \sim_p E'$. Equivalently, $\sim_p$ is closed under the rules for $\sim_p^1$: if the witness u of the rule π_1 differs from both binding atoms one appeals directly to π, and otherwise one may assume without loss of generality that $u = a$ and use theorem 41. ([Crole2012], Theorem 4.2.)*

Proof. □

Theorem 43 (Theorem 4.2: $\sim_p$ equals $\sim_p^1$)**.** *The relation $\sim_p$ of α-equivalence according to theorem 12 is identical to the relation $\sim_p^1$ of theorem 6. ([Crole2012], Theorem 4.2.)*

Proof. □

Proposition 44 (Proposition 4.3: variants of the side condition of π)**.** *Consider the variants $\sim_p^i$ of theorem 12 obtained by altering the side condition of the rule π:*

rule	*relation*	*condition*	$=\sim_p$
π	$\sim_p$	$z \not\sharp a, b, E, E'$	*yes*
π_1	$\sim_p^1$	$z \not\sharp E, E'$	*yes*
π_2	$\sim_p^2$	$z \not\sharp a, b$	*no*
π_3	$\sim_p^3$	*none*	*no*

The last column states for which i the relation $\sim_p^i$ is equal to $\sim_p$: the first two rows hold by theorem 43, while for $a \neq b$ the pairs $B([a]P(a,z)), B([b]P(b,a))$ and $B([a]P(b,a)), B([b]P(a,b))$ are respectively related by $\sim_p^2$ and by $\sim_p^3$ but are not α-equivalent, having different free atoms. ([Crole2012], Proposition 4.3.)

Proof. □

Lemma 45 ($\sim_r$ is contained in $\sim_p$)**.** *For all expressions E, E' we have $E \sim_r E' \implies E \sim_p E'$. Equivalently, $\sim_p$ is closed under the rules for $\sim_r$: the rules* ref, sym, trs *and* bcg *are theorem 15, theorem 13, theorem 17 and theorem 16,* pcg *is immediate, and the rule α is the instance theorem 36 of Lemma 6.5. ([Crole2012], Theorem 4.4.)*

Proof. □

Lemma 46 ($\sim_p$ is contained in $\sim_r$)**.** *For all expressions E, E' we have $E \sim_p E' \implies E \sim_r E'$. Equivalently, $\sim_r$ is closed under the rules for $\sim_p$; for the rule π one appeals to theorem 24 twice, to relate $(z\,a) \cdot E$ with $E\{z/a\}$ and $(z\,b) \cdot E'$ with $E'\{z/b\}$, and then uses* bcg*, two instances of α,* sym *and* trs*. ([Crole2012], Theorem 4.4.)*

Proof. □

Theorem 47 (Theorem 4.4: $\sim_p$ equals $\sim_r$)**.** *The relation $\sim_p$ of α-equivalence according to theorem 12 is identical to the relation $\sim_r$ of theorem 9. ([Crole2012], Theorem 4.4.)*

Proof. □

Lemma 48 ($\sim_{r\#}$ is contained in $\sim_p$)**.** *For all expressions E, E' we have $E \sim_{r\#} E' \implies E \sim_p E'$. The cases are exactly those of theorem 45, with the rule α replaced by $\alpha\#$; the latter is where the strengthening theorem 35 of the paper's inductive argument is needed, here in the form theorem 36. ([Crole2012], Theorem 4.5.)*

Proof. □

Lemma 49 ($\sim_p$ is contained in $\sim_{r\#}$)**.** *For all expressions E, E' we have $E \sim_p E' \implies E \sim_{r\#} E'$. The steps are exactly those of theorem 46, with theorem 26 used in place of theorem 24 and the two instances of α replaced by $\alpha\#$, which is legitimate because $z \not\triangleright E, E'$ implies $z \# E, E'$. ([Crole2012], Theorem 4.5.)*

Proof. □

Theorem 50 (Theorem 4.5: $\sim_p$ equals $\sim_{r\#}$)**.** *The relation $\sim_p$ of α-equivalence according to theorem 12 is identical to the relation $\sim_{r\#}$ of theorem 10. ([Crole2012], Theorem 4.5.)*

Proof. □

Theorem 51 (Theorem 4.6: $\sim_r$ equals $\sim_{r\#}$)**.** *The relation $\sim_r$ of α-equivalence according to theorem 9 is identical to the relation $\sim_{r\#}$ of theorem 10. ([Crole2012], Theorem 4.6.)*

Proof. Immediate from theorem 47 and theorem 50. □

Chapter 2

De Bruijn terms really do work

This chapter follows M. Norrish and R. Vestergaard, *Proof Pearl: de Bruijn Terms Really Do Work* (TPHOLs 2007). Results are numbered as in that paper.

2.1 Named abstraction (Section 2.1)

Definition 52 (The type Λ_α)**.** Λ_α is the quotient of the raw syntax Λ^{var} of named λ-terms by α-equivalence. We write $\mathsf{VAR}\, v$, $\mathsf{APP}\, M\, N$ and $\mathsf{LAM}\, v\, M$ for the operations on Λ_α induced by the three constructors of Λ^{var}. The operation LAM is not injective, so Λ_α is not a free algebra. ([Norrish2007a], Section 2.1.)

Definition 53 (Substitution on Λ_α)**.** Substitution $M[v := N]$ on Λ_α is induced by capture-avoiding substitution on raw terms, which respects α-equivalence in both term arguments. It has type $\Lambda_\alpha \times V \times \Lambda_\alpha \to \Lambda_\alpha$. ([Norrish2007a], Table 1, row λ^N_α.)

Definition 54 (β-reduction of λ^N_α)**.** The relation $\to_\beta$ on Λ_α is the least relation satisfying

$$\frac{}{\mathsf{APP}\,(\mathsf{LAM}\, v\, M)\, N \to_\beta M[v := N]} \qquad \frac{M \to_\beta N}{\mathsf{APP}\, M\, Z \to_\beta \mathsf{APP}\, N\, Z}$$

$$\frac{M \to_\beta N}{\mathsf{APP}\, Z\, M \to_\beta \mathsf{APP}\, Z\, N} \qquad \frac{M \to_\beta N}{\mathsf{LAM}\, v\, M \to_\beta \mathsf{LAM}\, v\, N}$$

It is defined on the quotient Λ_α of theorem 52 itself, using the substitution of theorem 53. ([Norrish2007a], Table 1, row λ^N_α.)

2.2 De Bruijn's anonymous abstraction (Section 2.2)

Definition 55 (de Bruijn terms)**.** The type dB of de Bruijn terms is the free algebra generated by

$$\mathsf{dB} \;\cong\; \mathbb{N} + \mathsf{dB} \times \mathsf{dB} + \mathsf{dB}$$

with constructors dV, dAPP and dABS for the three cases. ([Norrish2007a], equation (2).)

Definition 56 (Figure 2: lift)**.** The function lift adds one to all free indices of a term that are at least the given bound:

$$\begin{aligned}\mathsf{lift}\,(\mathsf{dV}\,i)\,k &= \text{if } i < k \text{ then } \mathsf{dV}\,i \text{ else } \mathsf{dV}\,(i+1)\\ \mathsf{lift}\,(\mathsf{dAPP}\,t\,u)\,k &= \mathsf{dAPP}\,(\mathsf{lift}\,t\,k)\,(\mathsf{lift}\,u\,k)\\ \mathsf{lift}\,(\mathsf{dABS}\,t)\,k &= \mathsf{dABS}\,(\mathsf{lift}\,t\,(k+1))\end{aligned}$$

([Norrish2007a], Figure 2, after Nipkow.)

Definition 57 (Figure 2: nsub)**.** Nipkow's substitution nsub substitutes for an index and at the same time decrements all larger free indices:

$$\begin{aligned}\mathsf{nsub}\,z\,k\,(\mathsf{dV}\,i) &= \text{if } k < i \text{ then } \mathsf{dV}\,(i-1) \text{ else if } i = k \text{ then } z \text{ else } \mathsf{dV}\,i\\ \mathsf{nsub}\,z\,k\,(\mathsf{dAPP}\,t\,u) &= \mathsf{dAPP}\,(\mathsf{nsub}\,z\,k\,t)\,(\mathsf{nsub}\,z\,k\,u)\\ \mathsf{nsub}\,z\,k\,(\mathsf{dABS}\,t) &= \mathsf{dABS}\,(\mathsf{nsub}\,(\mathsf{lift}\,z\,0)\,(k+1)\,t)\end{aligned}$$

([Norrish2007a], Figure 2, after Nipkow.)

Definition 58 (Definition 1: the relation $\to_d$)**.** The relation $\to_d$ on de Bruijn terms is the least relation satisfying

$$\frac{}{\mathsf{dAPP}\,(\mathsf{dABS}\,t)\,u \to_d \mathsf{nsub}\,u\,0\,t} \qquad \frac{t \to_d u}{\mathsf{dAPP}\,t\,z \to_d \mathsf{dAPP}\,u\,z}$$

$$\frac{t \to_d u}{\mathsf{dAPP}\,z\,t \to_d \mathsf{dAPP}\,z\,u} \qquad \frac{t \to_d u}{\mathsf{dABS}\,t \to_d \mathsf{dABS}\,u}$$

with nsub as in theorem 57. ([Norrish2007a], Definition 1, after Nipkow.)

2.3 An intermediate formalism (Section 4.1)

Definition 59 (Definition 4: a clean notion of substitution for dB)**.**

$$\begin{aligned}\mathsf{sub}\,z\,k\,(\mathsf{dV}\,i) &= \text{if } i = k \text{ then } z \text{ else } \mathsf{dV}\,i\\ \mathsf{sub}\,z\,k\,(\mathsf{dAPP}\,t\,u) &= \mathsf{dAPP}\,(\mathsf{sub}\,z\,k\,t)\,(\mathsf{sub}\,z\,k\,u)\\ \mathsf{sub}\,z\,k\,(\mathsf{dABS}\,t) &= \mathsf{dABS}\,(\mathsf{sub}\,(\mathsf{lift}\,z\,0)\,(k+1)\,t)\end{aligned}$$

Unlike nsub (theorem 57), it does not decrement the other free indices. ([Norrish2007a], Definition 4.)

Definition 60 (Free indices)**.** The finite set $\mathsf{dFV}(t)$ of free indices of a de Bruijn term t is defined by

$$\mathsf{dFV}(\mathsf{dV}\,i) = \{i\} \qquad \mathsf{dFV}(\mathsf{dAPP}\,t\,u) = \mathsf{dFV}(t) \cup \mathsf{dFV}(u) \qquad \mathsf{dFV}(\mathsf{dABS}\,t) = \{\, j \mid j+1 \in \mathsf{dFV}(t) \,\}$$

(Used in [Norrish2007a], Lemma 6 and Section 4.2.)

Definition 61 (Definition 5: abstraction over an index)**.**

$$\mathsf{dLAM}\,i\,t = \mathsf{dABS}\,(\mathsf{sub}\,(\mathsf{dV}\,0)\,(i+1)\,(\mathsf{lift}\,t\,0))$$

with sub as in theorem 59 and lift as in theorem 56. Like LAM on Λ_α, the function dLAM is not injective. ([Norrish2007a], Definition 5.)

Lemma 62 (Lemma 6, main induction)**.** *If $i \notin \mathsf{dFV}(t)$ then for every n there is a t_0 with $t = \mathsf{sub}\,(\mathsf{dV}\,n)\,i\,(\mathsf{lift}\,t_0\,n)$. ([Norrish2007a], proof of Lemma 6.)*

Proof. By structural induction on t. □

Lemma 63 (Lemma 6: dABS terms as dLAM terms)**.** *If i is not among the free indices of $\mathsf{dABS}\,t$, then there is a t_0 such that $\mathsf{dABS}\,t = \mathsf{dLAM}\,i\,t_0$. ([Norrish2007a], Lemma 6.)*

Proof. From theorem 62 with $n = 0$ and the index $i + 1$. □

Definition 64 (Definition 7: dB' β-reduction)**.** The relation $\to_{d'}$ on de Bruijn terms is the least relation satisfying

$$\frac{}{\mathsf{dAPP}\,(\mathsf{dLAM}\,i\,t)\,u \to_{d'} \mathsf{sub}\,u\,i\,t} \qquad \frac{t \to_{d'} u}{\mathsf{dAPP}\,t\,z \to_{d'} \mathsf{dAPP}\,u\,z}$$

$$\frac{t \to_{d'} u}{\mathsf{dAPP}\,z\,t \to_{d'} \mathsf{dAPP}\,z\,u} \qquad \frac{t \to_{d'} u}{\mathsf{dLAM}\,i\,t \to_{d'} \mathsf{dLAM}\,i\,u}$$

([Norrish2007a], Definition 7.)

2.4 Nominal reasoning with indices (Section 4.2)

Definition 65 (Definition 8: incrementing a permutation)**.** Write $\hat{i}$ for the atom with number i and $\hat{a}$ for the number of the atom a. For $g \in \mathbb{N}$ let ginc_g be the map on atoms that fixes a if $\hat{a} < g$ and sends it to the atom with number $\hat{a} + 1$ otherwise. For a finitely supported permutation π of atoms, $inc_g(\pi)$ is the permutation that fixes $\hat{g}$ and satisfies

$$inc_g(\pi)\,(\mathrm{ginc}_g\,a) = \mathrm{ginc}_g\,(\pi\,a)$$

for all atoms a. The paper's *inc* is inc_0. ([Norrish2007a], Definition 8 and the remark before Lemma 9. The paper defines *inc* on permutations represented as lists of pairs.)

Definition 66 (Definition 8: permutation for dB)**.** The action $\pi \cdot t$ of a finitely supported permutation π of atoms on a de Bruijn term t is

$$\pi \cdot (\mathsf{dV}\,i) = \mathsf{dV}\,\widehat{\pi(\hat{i})} \qquad \pi \cdot (\mathsf{dAPP}\,t\,u) = \mathsf{dAPP}\,(\pi \cdot t)\,(\pi \cdot u) \qquad \pi \cdot (\mathsf{dABS}\,t) = \mathsf{dABS}\,(inc_0(\pi) \cdot t)$$

with inc_0 as in theorem 65. ([Norrish2007a], Definition 8.)

Lemma 67 (The free indices support a term)**.** *If a permutation π fixes $\hat{i}$ for every $i \in \mathsf{dFV}(t)$, then $\pi \cdot t = t$. ([Norrish2007a], Section 4.2: the support of a de Bruijn term is its set of free indices converted into atoms.)*

Proof. By structural induction on t. □

Lemma 68 (Lemma 9: permutations and lift)**.** $\mathsf{lift}\,(\pi \cdot t)\,n = inc_n(\pi) \cdot (\mathsf{lift}\,t\,n)$. *([Norrish2007a], Lemma 9.)*

Proof. By induction on t. The dABS case needs $inc_0(inc_n\,\pi) = inc_{n+1}(inc_0\,\pi)$. □

Lemma 69 (Theorem 10, first part: sub has empty support)**.** $\pi \cdot (\mathsf{sub}\,t\,j\,u) = \mathsf{sub}\,(\pi \cdot t)\,\widehat{\pi(\hat{j})}\,(\pi \cdot u)$. *([Norrish2007a], Theorem 10.)*

Proof. By structural induction on u, using theorem 68. □

Lemma 70 (Theorem 10, second part: dLAM has empty support)**.** $\pi\cdot(\mathsf{dLAM}\,i\,t) = \mathsf{dLAM}\,\widehat{\pi(i)}\,(\pi\cdot t)$. *([Norrish2007a], Theorem 10.)*

Proof. Immediate from theorem 69 and theorem 68. □

Theorem 71 (Theorem 10: $\to_{d'}$ is equivariant)**.** $\pi\cdot t \to_{d'} \pi\cdot u \iff t \to_{d'} u$. *([Norrish2007a], Theorem 10.)*

Proof. The implication from right to left is a rule induction using theorem 69 and theorem 70. The converse follows because every permutation has an inverse. □

Lemma 72 (Free indices of dLAM)**.** $\mathsf{dFV}(\mathsf{dLAM}\,i\,t) = \mathsf{dFV}(t)\setminus\{i\}$. *([Norrish2007a], proof of Theorem 11, stated there as* $\mathrm{supp}(\mathsf{dLAM}\,i\,t) = \mathrm{supp}(t)\setminus\{\hat{i}\}$*.)*

Proof. By computing the free indices of $\mathsf{lift}\,t\,0$ and of a sub-substitution. □

Lemma 73 (Substitution under dLAM)**.** *If* $i \neq j$ *and* $i \notin \mathsf{dFV}(z)$ *then* $\mathsf{sub}\,z\,j\,(\mathsf{dLAM}\,i\,t) = \mathsf{dLAM}\,i\,(\mathsf{sub}\,z\,j\,t)$. *(Not numbered in [Norrish2007a]; it is* `sub_dLAM` *in the HOL4 sources.)*

Proof. Two substitutions for distinct indices commute when neither substituted term contains the other index free, and lift commutes with sub. □

Definition 74 (The map f on raw terms)**.** On raw named terms over atoms, the equations (5) of [Norrish2007a] define a function into de Bruijn terms by structural recursion:

$$f(v) = \mathsf{dV}\,\hat{v} \qquad f(M\,N) = \mathsf{dAPP}\,(f\,M)\,(f\,N) \qquad f(\lambda v.\,M) = \mathsf{dLAM}\,\hat{v}\,(f\,M)$$

with dLAM as in theorem 61.

Lemma 75 (f respects α-equivalence)**.** *If two raw terms* M *and* N *are* α*-equivalent then* $f(M) = f(N)$*, for* f *as in theorem 74. (Takes the place of the recursion principle for* Λ_α *used in [Norrish2007a], Theorem 11.)*

Proof. By rule induction, using theorem 70, theorem 72 and theorem 67. □

Definition 76 (The map $f\colon \Lambda_\alpha \to \mathsf{dB}$)**.** By theorem 75 the function of theorem 74 induces a function $f\colon \Lambda_\alpha \to \mathsf{dB}$. It satisfies the equations

$$f(\mathsf{VAR}\,v) = \mathsf{dV}\,\hat{v} \qquad f(\mathsf{APP}\,M\,N) = \mathsf{dAPP}\,(f\,M)\,(f\,N) \qquad f(\mathsf{LAM}\,v\,M) = \mathsf{dLAM}\,\hat{v}\,(f\,M)$$

([Norrish2007a], equation (5).)

Lemma 77 (f is injective up to α-equivalence)**.** *For raw terms* M *and* N*:* $f(M) = f(N)$ *if and only if* M *and* N *are* α*-equivalent. ([Norrish2007a], Theorem 11, injectivity.)*

Proof. One direction is theorem 75. The other is a structural induction whose abstraction case is theorem 86. □

Lemma 78 (f is surjective)**.** *Every de Bruijn term is* $f(M)$ *for some raw term* M*. ([Norrish2007a], Theorem 11, surjectivity.)*

Proof. By complete induction on the size of the de Bruijn term, using theorem 63 in the dABS case. □

Theorem 79 (Theorem 11: dB is in bijection with Λ_α)**.** *The function $f\colon \Lambda_\alpha \to \mathsf{dB}$ of theorem 76 is a bijection. ([Norrish2007a], Theorem 11.)*

Proof. From theorem 77 and theorem 78. □

Lemma 80 (f and substitution)**.** $f(M[v := N]) = \mathsf{sub}\,(f\,N)\,\hat{v}\,(f\,M)$ *for all* $M, N \in \Lambda_\alpha$. *([Norrish2007a], proof of Theorem 12.)*

Proof. By induction on the size of a representative of M, renaming the bound atom of an abstraction to a sufficiently fresh one and using theorem 73. □

Theorem 81 (Theorem 12)**.** $M \to_\beta N \iff f(M) \to_{d'} f(N)$ *for all* $M, N \in \Lambda_\alpha$. *([Norrish2007a], Theorem 12.)*

Proof. Two rule inductions, both relying on theorem 80. □

2.5 The homomorphism between dB and dB' (Section 4.3)

Lemma 82 (Lemma 13)**.** *If $n \le i$ then*

$$\mathsf{sub}\,u\,i\,t = \mathsf{nsub}\,u\,n\,(\mathsf{sub}\,(\mathsf{dV}\,n)\,(i+1)\,(\mathsf{lift}\,t\,n)).$$

([Norrish2007a], Lemma 13.)

Proof. By structural induction on t. □

Definition 83 (The lifting permutation)**.** For a lower bound ℓ and an upper bound m, the permutation *lift-pm*(ℓ, m) maps $\hat{\imath}$ to $\widehat{i+1}$ if $\ell \le i \le m$, maps $\widehat{m+1}$ back to $\hat{\ell}$, and fixes all other atoms. ([Norrish2007a], before Lemma 14.)

Lemma 84 (Lemma 14: lifting is a permutation)**.** *If n is at least as large as the largest free index in t, then*

$$\mathsf{lift}\,t\,m = \textit{lift-pm}(m, n) \cdot t.$$

([Norrish2007a], Lemma 14.)

Proof. By structural induction on t. The dABS case needs $\textit{inc}_0(\textit{lift-pm}(m, n)) = \textit{lift-pm}(m+1, n+1)$. □

Lemma 85 (Substitution of a fresh index is a permutation)**.** *If $i \notin \mathsf{dFV}(t)$ then* $\mathsf{sub}\,(\mathsf{dV}\,i)\,j\,t = (\hat{\imath}\ \hat{\jmath}) \cdot t$. *([Norrish2007a], Section 4.3, "The Nominal Aha!": a substitution of a fresh variable into a term is a permutation.)*

Proof. By structural induction on t, using that $\textit{inc}_0$ of the transposition $(\hat{\imath}\ \hat{\jmath})$ is the transposition of the atoms numbered $i+1$ and $j+1$. □

Lemma 86 (dLAM is injective up to a transposition)**.** $\mathsf{dLAM}\,i\,t = \mathsf{dLAM}\,j\,u$ *if and only if either $i = j$ and $t = u$, or $i \ne j$, $i \notin \mathsf{dFV}(u)$ and $t = (\hat{\imath}\ \hat{\jmath}) \cdot u$. (Not stated in [Norrish2007a]. It replaces the appeal to the recursion principle for Λ_α in the proof of Theorem 11.)*

Proof. Both sides are rewritten with theorem 85. The claim then follows from theorem 67, theorem 68 and the injectivity of lift. □

Theorem 87 (Theorem 15: abstraction congruences correspond)**.** *Let R be an equivariant relation on de Bruijn terms, that is $R\,(\pi \cdot t)\,(\pi \cdot u) \iff R\,t\,u$ for all π. Then*

$$(\forall t\,u.\ R\,t\,u \Rightarrow R\,(\mathsf{dABS}\,t)\,(\mathsf{dABS}\,u)) \iff (\forall i\,t\,u.\ R\,t\,u \Rightarrow R\,(\mathsf{dLAM}\,i\,t)\,(\mathsf{dLAM}\,i\,u)).$$

([Norrish2007a], Theorem 15.)

Proof. Immediate from theorem 84, theorem 85 and theorem 63. □

Lemma 88 (Lemma 16: $\rightarrow_d \subseteq \rightarrow_{d'}$)**.** $t \rightarrow_d u \implies t \rightarrow_{d'} u$. *([Norrish2007a], Lemma 16.)*

Proof. A rule induction. The redex case is theorem 82, and the dABS congruence case is theorem 87 applied to $\rightarrow_{d'}$, which is equivariant by theorem 71. □

Lemma 89 (Lemma 17)**.** $\pi \cdot (\mathsf{nsub}\,t\,i\,u) = \mathsf{nsub}\,(\pi \cdot t)\,i\,(inc_i(\pi) \cdot u)$. *([Norrish2007a], Lemma 17.)*

Proof. By structural induction on u. The dABS case uses theorem 68. □

Lemma 90 ($\rightarrow_d$ is equivariant)**.** $\pi \cdot t \rightarrow_d \pi \cdot u \iff t \rightarrow_d u$. *([Norrish2007a], after Lemma 17.)*

Proof. By rule induction, using theorem 89. □

Lemma 91 (Lemma 18: $\rightarrow_{d'} \subseteq \rightarrow_d$)**.** $t \rightarrow_{d'} u \implies t \rightarrow_d u$. *([Norrish2007a], Lemma 18.)*

Proof. A rule induction. The redex case is theorem 82, and the dLAM congruence case is theorem 87 applied to $\rightarrow_d$, which is equivariant by theorem 90. □

Theorem 92 (Theorem 19: the relations $\rightarrow_d$ and $\rightarrow_{d'}$ coincide)**.** $t \rightarrow_{d'} u \iff t \rightarrow_d u$. *([Norrish2007a], Theorem 19.)*

Proof. Combining theorem 88 and theorem 91. □

Lemma 93 (Theorem 20, homomorphism property)**.** $M \rightarrow_\beta N \iff f(M) \rightarrow_d f(N)$ *for all* $M, N \in \Lambda_\alpha$. *([Norrish2007a], Theorem 20.)*

Proof. From theorem 81 and theorem 92. □

Theorem 94 (Theorem 20: isomorphism)**.** *The types* dB *and* Λ_α*, with their respective reduction relations* $\rightarrow_d$ *and* $\rightarrow_\beta$*, are isomorphic: there is a bijection* $f : \Lambda_\alpha \rightarrow \mathsf{dB}$ *with*

$$M \rightarrow_\beta N \iff f(M) \rightarrow_d f(N).$$

([Norrish2007a], Theorem 20.)

Proof. The bijection is that of theorem 79; it is a homomorphism by theorem 93. □

2.6 Eta-reduction (Section 4.4)

Definition 95 (η-reduction on Λ_α)**.** The relation $\to_\eta$ on Λ_α is the least relation closed under the three congruence rules of theorem 54 and the contraction rule

$$\frac{v \notin \mathrm{FV}(M)}{\mathsf{LAM}\, v\, (\mathsf{APP}\, M\, (\mathsf{VAR}\, v)) \to_\eta M}$$

([Norrish2007a], Section 4.4.)

Definition 96 (Nipkow's η-reduction $\to_{ne}$)**.** The relation $\to_{ne}$ on de Bruijn terms is the least relation closed under the congruence rules for dAPP and dABS and the redex rule

$$\frac{0 \notin \mathsf{dFV}(t)}{\mathsf{dABS}\, (\mathsf{dAPP}\, t\, (\mathsf{dV}\, 0)) \to_{ne} \mathsf{nsub}\, u\, 0\, t}$$

in which the choice of u is immaterial. ([Norrish2007a], Section 4.4.)

Definition 97 (The η-reduction $\to_e$)**.** The relation $\to_e$ on de Bruijn terms is the least relation closed under the congruence rules for dAPP and dLAM and the contraction rule

$$\frac{i \notin \mathsf{dFV}(t)}{\mathsf{dLAM}\, i\, (\mathsf{dAPP}\, t\, (\mathsf{dV}\, i)) \to_e t}$$

([Norrish2007a], Section 4.4.)

Lemma 98 (nsub after lift)**.** $\mathsf{nsub}\, u\, n\, (\mathsf{lift}\, t\, n) = t$. *([Norrish2007a], Section 4.4, a lemma of Nipkow.)*

Proof. By structural induction on t. □

Lemma 99 (lift after nsub)**.** *If* $n \notin \mathsf{dFV}(t)$ *then* $\mathsf{lift}\, (\mathsf{nsub}\, u\, n\, t)\, n = t$. *([Norrish2007a], Section 4.4.)*

Proof. By structural induction on t. □

Theorem 100 (The relations $\to_e$ and $\to_{ne}$ coincide)**.** $t \to_e u \iff t \to_{ne} u$. *([Norrish2007a], Section 4.4.)*

Proof. The redex rules correspond by theorem 98 and theorem 99; the abstraction congruences correspond by theorem 87, since both relations are equivariant. □

Theorem 101 (η-reduction corresponds along f)**.** $M \to_\eta N \iff f(M) \to_{ne} f(N)$ *for all* $M, N \in \Lambda_\alpha$, *with* $\to_\eta$ *as in theorem 95. ([Norrish2007a], Section 4.4.)*

Proof. The relation $\to_\eta$ corresponds to $\to_e$ along f as in theorem 81; then apply theorem 100. □

2.7 Using the isomorphism

Corollary 102 (Church–Rosser for λ^N_α)**.** *The relation* $\to_\beta$ *on* Λ_α *is confluent. (Not a numbered result of [Norrish2007a]; it illustrates the use of Theorem 20.)*

Proof. Confluence of $\to_d$ on de Bruijn terms is transported along the isomorphism of theorem 94. □

Corollary 103 (Uniqueness of β-normal forms in Λ_α)**.** *If* M *reduces in several steps to* N_1 *and to* N_2, *and both are* β*-normal, then* $N_1 = N_2$. *(Not a result of [Norrish2007a].)*

Proof. From theorem 102. □